Hierarchical Feature Progressive Alignment Network for Remote Sensing Image Scene Classification in Multitarget Domain Adaptation

Wang Miao, Wenqi Han, Jie Geng, *Member, IEEE*, and Wen Jiang

***Abstract*— Multitarget domain adaptation (MTDA) presents a formidable challenge in remote sensing image scene classification (RSICS), where the objective is to transfer knowledge from a labeled source domain to several unlabeled target domains. Compared to single-source-single-target domain adaptation (S3TDA), MTDA is inherently more complex due to domain shifts among multiple target domains. Directly merging the unique features of multitarget domains can result in corrupted information and poor classification performance. To address these challenges, we propose a hierarchical feature progressive alignment network (HFPAN) for RSICS in MTDA. First, our method introduces a fine-grained and contextual information extraction (FCIE) network to extract the global-local correlation in remote sensing (RS) images. Second, we construct a hierarchical feature embedding (HFE) framework that maintains hierarchical inter, intra constraints for the extracted features. Finally, we perform an alignment process for the constructed hierarchical features to minimize the differences in MTDA, progressing from coarse to fine granularity. To evaluate the efficacy of our proposed method, we conducted several cross-domain scene classification experiments on five public datasets. These experiments demonstrate the novelty of our approach and its ability to achieve improved classification performance.**

***Index Terms*— Domain adaptation (DA), multitarget DA (MTDA), remote sensing (RS) image, scene classification.**

I. Introduction

DEEP learning methods have exhibited remarkable achievements in various remote sensing (RS) tasks, as evidenced by extensive research [1], [2], [3], [4], [5], [6], [7], [8], [9]. However, effective representations in these methods heavily depend on manual annotations for a specific distribution, commonly referred to as the source domain [10], [11], [12], [13], [14], [15]. Directly applying models trained on source domains to target domains inevitably leads to poor classification performance due to differences in data distributions and variations caused by factors such as sensor differences and surface environments [16], [17], [18]. RS image (RSI) from different datasets with the same annotations exhibit significant disparities in spectral distributions [19], [20], [21], including variations in textures, colors, and other characteristics [3], [4], [16]. Although fine-tuning pretrained networks on task-specific datasets is a common solution, the expensive cost of label collection often renders this approach impractical [3], [22], [23], [24], [25].

Recent studies have increasingly focused on enhancing the applicability of deep learning techniques to RS imagery [26], [27], [28], [29], [30]. For example, Zhang et al. [26] introduced the structural optimization transmission network (SOT-Net), which enhances land-cover classification accuracy by utilizing hyperspectral (HSI) and LiDAR data. Wang et al. [30] developed the representation-enhanced status replay network (RSRNet), employing dual augmentation and a status replay strategy to mitigate representation bias and fortify classifier stability. Additionally, the latest advancements in deep learning have significantly enhanced scene classification in RS imagery [31], [32], [33]. The multistage self-guided separation network (MGSNet) proposed by Wang et al. [31] improves discrimination between similar targets and differing backgrounds by integrating background context and applying contrastive regularization. Zhao et al. [32] presented the efficient multisample contrastive network (EMSCNet), which utilizes a dynamic dictionary with momentum updates for mining data pairs, enhancing knowledge integration from multiple samples. Furthermore, Yang et al. [33] proposed the semantic-aware graph network (SAGN), comprising a dense feature pyramid network (DFPN), an adaptive semantic analysis module (ASAM), a dynamic graph feature update module, and a scene decision module (SDM). These components collaboratively extract multiscale information, analyze semantic diversity, uncover unstructured relationships, and render decisions for high-resolution RS scene classification.

Single-source-single-target domain adaptation (S3TDA) methods [20], [34], [35], [36], [37] focus on adapting a discriminative model trained on a labeled source domain to an unlabeled target domain by minimizing the distribution

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difference between them. These methods can be categorized into two types: MMD-based [20], [34], [35] and adversarial training-based [36], [37]. MMD-based methods minimize the marginal distribution discrepancy using the MMD metric, while adversarial training-based methods align the distributions through adversarial learning. However, these methods are limited to a single source-target domain pair and unsuitable for the MTDA task involving multiple target domains. The traditional S3TDA setting is inadequate for real-world scenarios in RS images, where different distributions are collected under various conditions.

To address these limitations, other deep domain adaptation (DA) methods have emerged as successful approaches in various RS applications [16], [18], [19], [38]. Zhu et al. [8] introduced the attention-based multiscale residual adaptation network (AMRAN) for single-target DA task, which leverages both marginal and conditional distributions to comprehensively align the source and target domains. Zheng et al. [3] tackled the challenge of partial DA in RSISC by introducing a progressive auxiliary domain that mitigates negative transfer effects resulting from outlier classes. They enhance the domain adversarial neural network (DANN) with multiple weights to improve domain confusion. Xu et al. [39] presented a practical solution for universal DA in RSISC, which does not require prior knowledge of the label sets. Their method introduced a novel approach that promotes adaptation in the discovered shared label set, even in the absence of source data. Huang et al. [40] utilized bidirectional sample-class alignment (BSCA), a novel technique for semi-supervised DA in RSISC. BSCA integrates unsupervised and supervised alignment strategies to effectively reduce domain shift.

MTDA has gained attention as it simultaneously adapts a single-source domain to multitarget domains. The MTDA setting is more complex than the S3TDA setting due to domain shifts among different target domains and each source-target pair. Extending prior S3TDA methods to MTDA can be approached in two ways. The first direction [20], [37] trains a model with a feature extractor and a single classifier independently on each source-target pair. However, this approach is impractical as it requires retraining the model for each new target domain and overlooks the correlation among multiple target domains. The second direction [19] utilizes data grouping techniques to merge all target domains into a combined target domain. The model is trained on the labeled source samples and the combined target samples using existing S3TDA methods. This method overcomes the limitations of the first approach, allowing the model to learn representations from both the source and multiple target domains simultaneously. Additionally, To address the challenge in MTDA, Zheng et al. [19] introduced TSAN, a two-stage adaptation network (TSAN), to address this problem. Similarly, Ngo et al. [41] proposed EHACA, a new method that follows an easy-to-hard adaptation structure. Its goal is to capture the semantic information in each target domain and establish connections between the source and target domains, as well as among different target domains. However, while achieving excellent performance, current cross-domain methods fail to consider the structural aspects of features between multiple domains, relying more on data enhancement techniques.

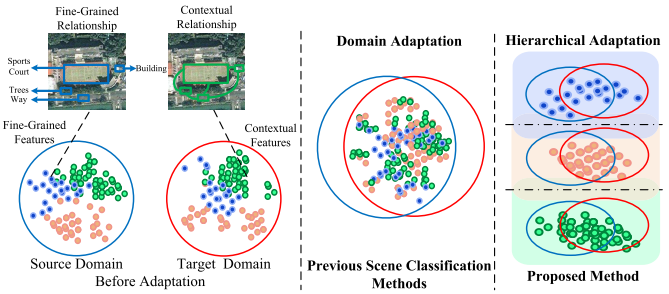


Fig. 1. Illustration of the difference between the current DA methods and our proposed approach. Existing approaches primarily concentrate on aligning features directly at the depth feature level.

Multiple research studies have demonstrated that RSI exhibits intricate contextual relationships and possesses various local features, including texture and color [42], [43], [44], [45], [46]. However, as shown in Fig. 1, existing approaches have primarily focused on aligning features at the depth level, leading to overfitting specific features and limited alignment effectiveness. Additionally, many studies have aligned features with the same physical meaning, overlooking the potential confusion within the feature space. To address these limitations, this article introduces a novel approach for mixed domain feature alignment in RS image characteristics, which involves constructing a fine-grained hierarchical feature space progressing from coarse-grained to fine-grained representation.

In this article, we propose a novel hierarchical feature progressive alignment method for RSISC in MTDA. The proposed HFPAN consists of three main components: a fine-grained and contextual information extraction (FCIE) network, a hierarchical feature embedding (HFE) framework for fine-grained and contextual features, and a hierarchical feature progressive alignment loss function. The key contributions of this work are as follows.

1) The hierarchical feature progressive alignment network (HFPAN) is proposed as a solution to overcome the challenges in multitarget DA (MTDA). HFPAN gradually aligns fine-grained features with complex contextual relationships, transitioning from coarse to fine levels of representation. This progressive alignment strategy enables the network to adapt stably to the variations of each domain, leading to stable and effective adaptation.
2) A HFE framework is constructed to maintain interclass and intraclass feature embeddings in MTDA. The HFE framework leverages the strengths of convolutional neural networks (CNNs) and multiscale transformer (MT) architectures for MTDA tasks. It forms a fine-grained feature space and avoids overfitting the model to certain types of features during the alignment process.
3) Building upon the constructed hierarchical feature space, we propose a hierarchical feature progressive alignment loss function that prioritizes the alignment of complex contextual features before fine-grained features. It effectively minimizes confusion between features during the

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alignment process by distinguishing them from existing cross-domain alignment methods. Moreover, it lessens the tendency of the network to overfit particular features within individual domains. The strategy ensures that the network incrementally adjusts its focus, moving from broad contextual alignments to more detailed ones, improving the DA's stability and efficacy.

The structure of our article is outlined as follows: Section II provides a concise overview of related works. Section III presents the details of our proposed HFPAN. The comparative results and experimental analyses are discussed in Section IV. Finally, Section V summarizes the conclusions derived from our study.

II. Related Work

A. DA in Computer Vision

DA methods have garnered significant research attention in various fields [47], [48], [49], [50]. Recent advancements in this field have introduced innovative methods to tackle the challenges of DA. For instance, Zhu et al. [47] proposed PatchMix, a module based on the ViT architecture that facilitates the construction of intermediate domain probability distributions. PatchMix leverages game-theoretical models to sample patches from source and target domains, optimizing cross-entropy and semi-supervised mixup losses. In the context of open-universe visual DA (OUVDA), Zara et al. [48] utilized pretrained language and vision models (CLIPs) known for their rich representation and zero-shot recognition capabilities. Their approach harnesses the strengths of CLIP for OUVDA. To address domain shift in cross-domain classification, Kang et al. [49] propose the contrastive adaptation network (CAN) with the novel metric of contrastive domain discrepancy (CDD). The CAN explicitly models both intraclass and interclass domain discrepancies, making it applicable to single-source and multisource DA scenarios. In the realm of unsupervised DA (UDA), Carlucci et al. [50] introduced a novel approach that incorporates domain alignment layers. To align the feature distributions between the source and target domains, these layers independently learn the appropriate degree of feature alignment at different levels within the deep network. These advancements contribute significantly to addressing the challenges of domain shift, demonstrating improved performance and representation alignment [50], [51].

B. DA in RS

The objective of UDA in RSI is to enhance generalization from a labeled source domain to an unlabeled target domain [16], [19]. Transfer learning, which leverages knowledge from pretrained models to train other models, plays a vital role in cross-domain RSISC tasks [38]. For instance, Zheng et al. [19] proposed a novel approach for UDA in optical RSISC and retrieval. Their method leverages self-supervised learning and consistent self-training to address the challenge of adapting a single source domain to multiple mixed target domains in RS applications. The algorithm, known as the TSAN, consists of two stages. In the first stage, adversarial learning is employed to confuse the classifier and make distinguishing between the source domain and the mixed multiple target domains difficult. In the second stage, the network learns the intrinsic features of each target domain by partitioning the mixed multiple target domains into multiple pseudo-domain labels using self-supervised learning. These two stages are combined into an iterative procedure that optimizes the model for adaptation to the mixed target domains. Furthermore, Zheng et al. [16] proposed the DA via task-specific classifier network (DATSNET) to address the domain shift problem in RSISC. To align the feature distributions between the source and target domains, the framework integrates task-specific classifiers and minimax discrepancy. This approach focuses on capturing domain-invariant features and improving classification performance in RSISC tasks.

III. Methodology

A. Overview

This article proposes a hierarchical progressive feature alignment network (HFPAN) for MTDA. MTDA endeavors to predict labels for multitarget domains by leveraging information from a solitary labeled source domain alongside unlabeled target domains. In the MTDA setting, the labeled training dataset consists of pairs of source images x_S^i and their associated labels y_S^i. Additionally, there are multiple unlabeled target domains T_j, each with its own set of unlabeled samples $x_{T_j}^i$, i denotes the ith sample, j denotes the jth target domain, $j = 1, 2, \ldots, N$, $i = 1, 2, \ldots, n$. The objective is to predict the labels of the target domains, assuming a shared label space $y_S = y_{T_j}$ for all domains. It is also assumed that the distributions of the target domains are different from each other and different from the source domain.

The HFPAN framework is specifically designed for RSI. It includes three main modules: the FCIE network, the HFE framework for fine-grained and contextual features, and the hierarchical feature progressive alignment loss function. Fig. 2 visually represents the HFPAN framework, illustrating the three main modules. The MTDA setting is more complex than the S3TDA setting due to domain shifts among different target domains and each source-target pair, leading to overfitting specific features and limited alignment effectiveness. Our hierarchical approach to learning is advantageous for MTDA, progressing from general to specific and from less complex to more complex features. It avoids the drawbacks of early fine-grained feature alignment, which can be detrimental due to the substantial variability of details across different domains. This step-by-step strategy permits the network to gradually internalize each domain's unique characteristics, leading to a more stable and effective DA process.

B. Fine-Grained and Contextual Information Extraction

Several research studies have demonstrated that RSI exhibits intricate contextual relationships and local features, including texture and color [42], [43], [44], [45], [46]. In this article, we propose the FCIE framework, specifically designed to address the challenging task of contextual information and fine-grained feature extraction in MTDA. The FCIE, depicted in Fig. 3, effectively captures global and local relations,

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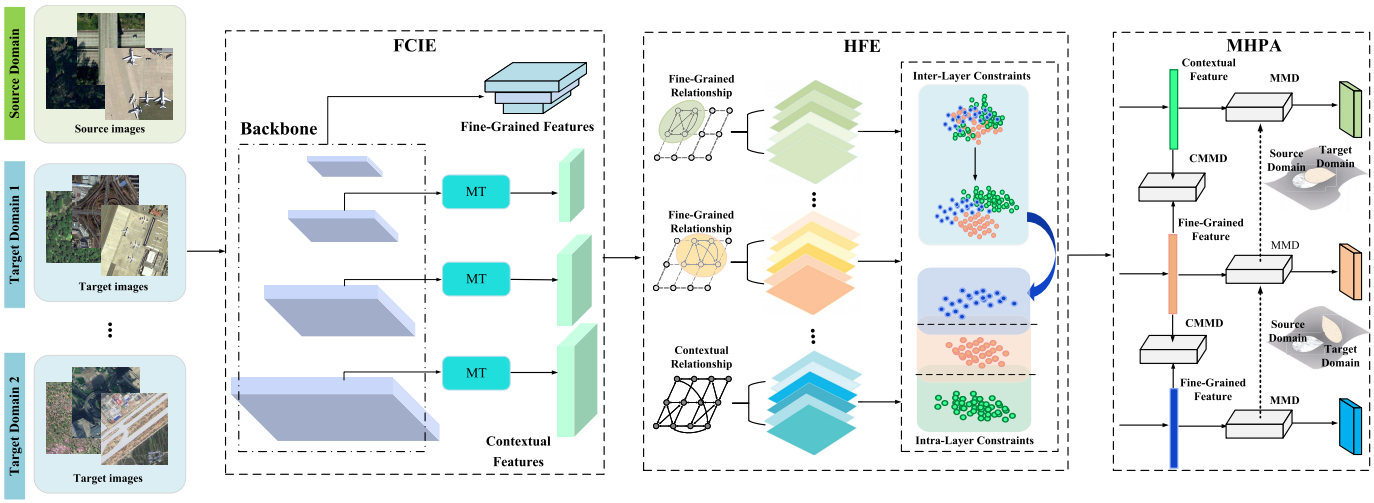


Fig. 2. Overall framework of the proposed HFPAN. It involves gradually aligning hierarchical features, considering the complex contextual and fine-grained relationships. The network architecture transitions from coarse to fine levels of representation, enabling effective feature alignment throughout the learning process.

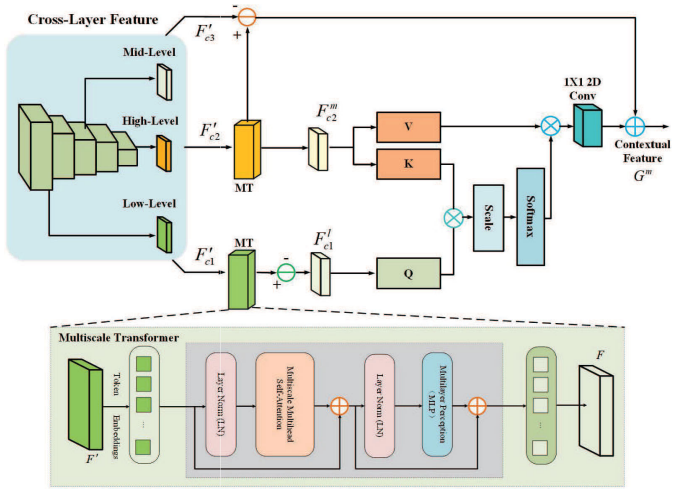


Fig. 3. Framework of the proposed FCIE.

enabling accurate classification of fine-grained species by embedding long-term dependencies.

The FCIE framework combines CNN and MT architectures to achieve the task of MTDA. It introduces a part-aware embedding that enhances the long-term dependencies of conventional CNNs. Fig. 3 illustrates our proposed feature extraction approach for MTDA. The FCIE incorporates two main ideas: integrating transformers and CNNs to establish contextual dependencies for capturing long-range dependencies and regularizing the network embedding to be aware of critical local discriminative regions in visually similar samples. We introduce one global relation transformation G_M and three local relation transformations F_c to capture global and regional relationships. These relation transformations acquire essential attention features by constructing densely connected regions during the process of gradient descent. The global relation transformation captures the global relationship between different part regions, while the local relation transformations capture the local relationships between adjacent regions.

In the FCIE framework, we utilize ResNet as the CNN in the proposed model. Various residual blocks contribute to comprehensive feature representations by capturing multilevel convolutional features. High-level convolutional features convey semantic information, whereas low-level convolutional features convey low-level information such as color, texture, and shape. To obtain more informative features of RSI, we employ the outputs of the last three residual blocks as the low-, mid-, and high-level convolutional features, denoted as $F^S_{c1,2,3} = \mathrm{CNN}(x^i_S)$ and $F^T_{c1,2,3} = \mathrm{CNN}(x^i_{T_j})$, respectively. These multilevel convolutional features of the source image x^i_S and unlabeled target sample $x^i_{T_j}$ contain rich local information of RSI, providing both local structural information and high-level feature representations. Then, we develop the efficient MT to capture long-range dependencies of RSI. The specific process is shown in Fig. 3. Therefore, the obtained features $F_{c1} = \mathrm{MT}(F_1)$, $F_{c2} = \mathrm{MT}(F_2)$, and $F_{c3} = \mathrm{MT}(F_3)$ contain abundant long-range information of RSI.

The integration of features at different levels is crucial for our proposed approach. We introduce a cross-level attention mechanism to capture the long-range correlation between features G_M with varying spatial scales. The cross-level attention mechanism effectively utilizes valuable semantic information alongside the diverse shallow messages found in low-level and intermediate-level features F^S_c and F^T_c. We apply average pooling and 1×1 convolution operations to reduce the spatial size of low- and mid-level features and unify the number of channels. Subsequently, we generate new features F^T_{c1} and F^T_{c2} with the same spatial size as the high-level features F^T_{c3}. We employ learnable parameters W_Q, W_K, and W_V to project $F^l_1 = F_{c1}$ and $F^m_2 = F_{c2}$ into the spaces of the queries (Q) and the keys (K/V), respectively. These steps can be expressed as follows:

$$Q = F^l_{c1} W^Q, K = F^m_{c2} W^K, V = F^m_{c2} W^V. \tag{1}$$

Next, we utilize the dot product and softmax function to calculate the attention weights between Q and K, resulting in the generation of new features. These new features are then

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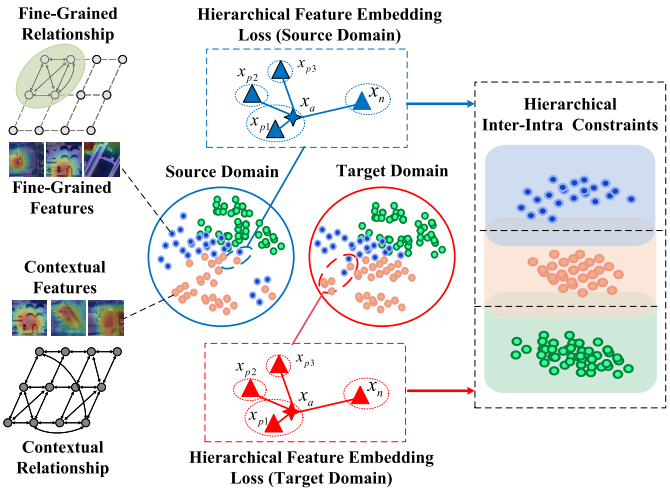


Fig. 4. Framework of the proposed HFE loss in MTDA.

added to $F_3 = F_{c3}$, resulting in the final aggregated feature G_M that encapsulates multilevel information. To sum up, the process can be represented as follows:

$$G_M = F_{c3}{}' + \text{soft max}\left(\frac{\left(F_{c1}^{l} W^{Q}\right)\left(F_{c2}^{m} W^{K}\right)^{T}}{\sqrt{d_k}}\right)\left(F_{c2}^{m} W^{V}\right). \tag{2}$$

The equation incorporates a binary mask represented by ε and the element-wise multiplication operation denoted by $\odot$. Additionally, the equation has a scaling factor of dk. While the generation of F_{c1}, F_{c2}, and F_{c3} in self-attention is based on single-level features, the summation features in cross-level attention are derived from features at different levels. As a result, the aggregated contextual feature G_M captures a wider range of granularity information from RSI in the MTDA task, providing a more comprehensive and diverse representation.

C. Hierarchical Feature Embedding

Existing RS image analysis approaches have primarily focused on aligning features directly at the depth feature level. However, it often leads to overfitting specific features and limited alignment effectiveness [52]. Furthermore, many studies have aligned features with the same physical meaning, disregarding the potential confusion within the feature space [42], [43], [44], [45], [46]. To overcome these limitations, this article introduces a novel approach that constructs a fine-grained hierarchical feature space for mixed domain feature alignment in RS image characteristics. It progresses from a coarse-grained to a fine-grained representation, allowing for more effective alignment.

Fig. 4 illustrates the proposed HFE Loss, which addresses the redundancy and duplication of features in the feature space. The objective of the HFE Loss is to separate the features before asymptotic alignment. This is achieved by controlling the nearest negative sample distance of different feature levels to be greater than the farthest positive sample distance of the same level. The HFE Loss is designed to work with HFE for fine-grained and contextual features. Specifically, it compares the anchor sample feature $x_{i,h}^{F_C}$ with the features of a positive sample $x_{i,h}^{G_M}$ and a negative sample $x_{i,h}^{F_C}$ at different level, with the interclass margin denoted as α_1 and α_2. The loss function is defined as follows:

$$L_{\text{HFE-F}} = \frac{1}{N}\sum_{i=1}^{N}\sum_{h=1}^{H}\left[d\left(x_{ih,a}^{F_C}, x_{ih,p3}^{F_C}\right) - d\left(x_{ih,a}^{F_C}, x_{ih,n1}^{G_M}\right) + \alpha 1\right] \tag{3}$$

$$L_{\text{HFE-G}} = \frac{1}{N}\sum_{i=1}^{N}\sum_{h=1}^{H}\left[d\left(x_{ih,a}^{F_C}, x_{ih,p1}^{F_C}\right) - d\left(x_{ih,a}^{G_M}, x_{ih,p2}^{F_C}\right) + \alpha 2\right]. \tag{4}$$

Here, h denotes the feature of the hth level, $d(x_{i,h}^{a}, x_{i,h}^{p1}) < d(x_{i,h}^{a}, x_{i,h}^{p2}) < d(x_{i,h}^{a}, x_{i,h}^{p3}) < d(x_{i,h}^{a}, x_{i,h}^{n})$, $x_{i,h}^{n1} = \arg\min_{x_{i,h}^{n}} d(x_{i,h}^{a}, x_{i,h}^{n})$, $x_{i,h}^{p1} = \arg\min_{x_{i,h}^{p}} d(x_{i,h}^{a}, x_{i,h}^{p})$, $x_{i,h}^{p2} = \arg\min_{x_{i,h}^{p}} d(x_{i,h}^{a}, x_{i,h}^{p})$, and $x_{i,h}^{p3} = \arg\max_{x_{i,h}^{p}} d(x_{i,h}^{a}, x_{i,h}^{p})$. To select triplets, we adopt the hard batch mode; suppose $x_{i,h}^{a}$ is the anchor sample, $x_{i,h}^{p1}$, $x_{i,h}^{p2}$, and $x_{i,h}^{p3}$ are the positive samples at the level of fine-grained feature F_C, and $x_{i,h}^{n}$ is the negative sample at the level of contextual feature G_M; $x_{i,h}^{p1}$ denotes the sample with the farthest distance at the same level, and $x_{i,h}^{p2}$ denotes the sample with the closest distance at the same level. It involves choosing the closest negative sample to the anchor within each batch as the hard negative sample $x_{i,h}^{n1}$, and the farthest positive sample as the hard positive sample $x_{i,h}^{p3}$.

Although the hierarchical embedding interclass loss is applied, the features within each class may still be mixed up. We incorporate RSI to mitigate this issue to create fine-grained intraclass feature embeddings. The intraclass feature embedding ensures that features of the same hierarchical level and class are closely grouped, enhancing the embedding's resilience to scene variation. We introduce a hard hierarchical embedding intraclass loss, where the closest positive sample to the anchor with different hierarchical features, but the same hierarchical feature replaces the hard negative sample, and the farthest positive sample with the same hierarchical feature replaces the hard positive sample. The intraclass loss is formulated as follows:

$$L_{\text{BR}} = \frac{1}{N}\sum_{i=1}^{N}\sum_{h=1}^{H}\left[\alpha 3 - d\left(x_{ih}^{a}, x_{ih}^{n}\right)\right]. \tag{5}$$

To ensure discriminative intraclass feature embeddings that consider both the difference and the absolute value of distances between samples, we introduce Boundary Regularization (BR). BR enforces a constraint that ensures the distance between the anchor and the negative sample is larger than a certain absolute distance $\alpha 3$ while simultaneously pulling the positive sample $x_{i,h}^{p2}$ away from the anchor relative to $x_{i,h}^{p1}$. This constraint helps create more discriminative intraclass feature embeddings. The BR is incorporated into the loss function, and the smaller margin $\alpha 3$ indirectly interacts with the interclass boundary.

The HFE can be expressed as follows:

$$L_{\text{HFE}} = \varepsilon L_{\text{HFE-F}} + \phi L_{\text{HFE-G}} + L_{\text{BR}} \tag{6}$$

where ε and ϕ are the balance factors used to control the impact of $L_{\text{HFE-F}}$ and $L_{\text{HFE-G}}$. The HFE Loss combines

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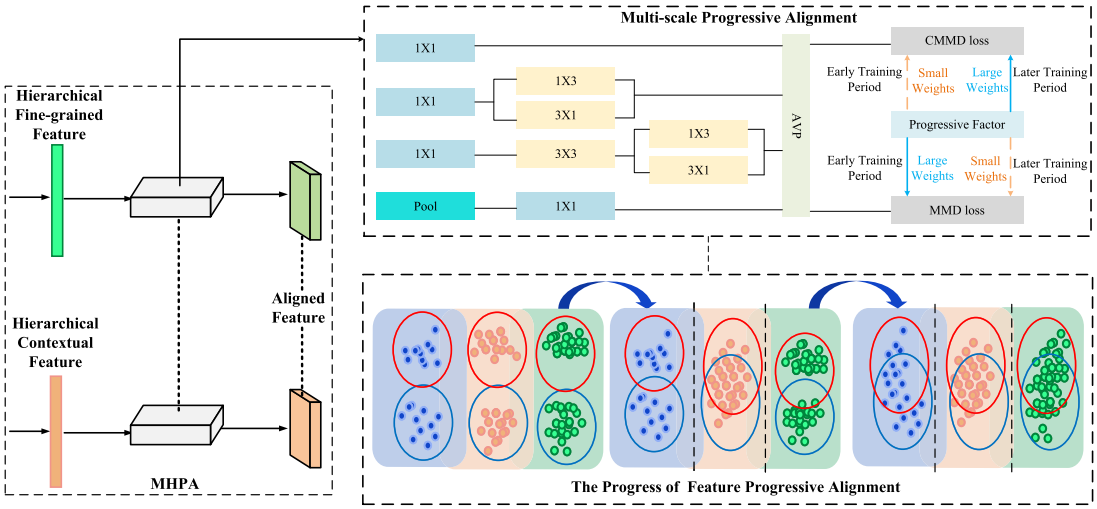


Fig. 5. Illustration of the proposed MHPA Loss in MTDA. We utilize a multiscale adaptation module to extract and align features across multiple scales to enhance distribution alignment.

interclass and intraclass losses. By integrating fine-grained and contextual features of RSI, it constructs a HFE that maintains multiple relative relationships between samples. Then, it enables the embedding to capture fine-grained details and contextual information, resulting in more discriminative and informative representations.

D. Multiscale Hierarchical Progressive Alignment

To reduce feature confusion in MTDA, building upon the constructed hierarchical feature space, we present a multiscale hierarchical progressive alignment (MHPA) approach for RSI that considers the human vision's tendency to perceive structure and frame before focusing on details. The constructed feature space follows a progression where contextual features are aligned first, followed by fine-grained features. This construction strategy mitigates confusion between fine-grained and contextual features, preventing network overfitting to specific features during progressive learning. It enables multilevel alignment, from coarse to fine-grained and global to local, considering different levels of fine-grained features and contextual features.

Specifically, MHPA aims to minimize the discrepancy between marginal distributions of contextual features and the discrepancy of conditional distributions of fine-grained features between the multidomains. Drawing inspiration from the multiscale inception module, we leverage multiscale representations that capture different receptive fields and more information compared to single-scale representations. By incorporating the conditional maximum mean discrepancy (CMMD) loss in each substructure, our multiscale adaptation module minimizes the conditional distribution discrepancies at each scale, ensuring greater conditional invariance in the extracted features. The attention module enhances the feature extraction's invariance. The overall structure of our proposed method is as depicted in Fig. 5. To simplify the formulas, we define $\mathrm{MH}(.) = \mathrm{MHPA}(\mathrm{FCIE}(x))$ as the features obtained after the multiscale structures, and $y = \mathrm{softmax}(\mathrm{MH}(\mathrm{FCIE}(x)))$ as the predicted labels.

In the context of different domains, the differences in fine-grained local features tend to be larger compared to global contextual features. To address this, we introduce a progressive factor that gradually reduces the domain differences from coarse to fine-grained levels. As shown in Fig. 5, for the global marginal distribution divergence $\mathbb{D}_G(X_S, X_T)$, we employ maximum mean discrepancy (MMD) to estimate the distance between the cross-domain distributions. For the local marginal distribution divergence $\mathbb{D}_L(X_S, X_T)$, we employ CMMD to estimate the distance between the cross-domain distributions. The global and local marginal distribution divergence $\mathbb{D}_G(X_S, X_T)$ and $\mathbb{D}_L(X_S, X_T)$ are defined as follows:

$$\mathrm{MMD}_{D_G}(X_S, X_T) = \left\| \frac{1}{n_1}\sum_{i=1}^{n_1} \beta^T h(\mathrm{MH}(G))_i - \frac{1}{n_2}\sum_{j=1}^{n_2} \beta^T h(\mathrm{MH}(G))_{j+n_1} \right\|_H^2 \tag{7}$$

$$\mathrm{CMMD}_{D_L}(X_S, X_T) = \left\| \frac{1}{n_1}\sum_{i=1}^{n_1} \beta^T h(\mathrm{MH}(F))_i - \frac{1}{n_2}\sum_{j=1}^{n_2} \beta^T h(\mathrm{MH}(F))_{j+n_1} \right\|_H^2 \tag{8}$$

where h_i is the ith column vector of the kernel matrix K, and β is the marginal matrix of MMD. Based on the aforementioned multiscale MMD and CMMD losses, we introduce a progressive factor that increases with the number of iterations, enabling the model to assign a higher weight to the alignment loss of contextual features during the early stages of training while assigning a lower weight to the alignment loss of fine-grained features. In the later stages of training, the weight assigned to the alignment loss of fine-grained features increases, while the weight assigned to the alignment loss of contextual features decreases. Then the proposed multiscale progressive alignment loss function can be written as

$$L_{\mathrm{PA}} = (1-\alpha)\cdot d_{\mathrm{MMD}}(F(S), F(T)) + \alpha \cdot \sum_{i=1}^{r} d_{\mathrm{CMMD}}(G(S), G(T)) \tag{9}$$

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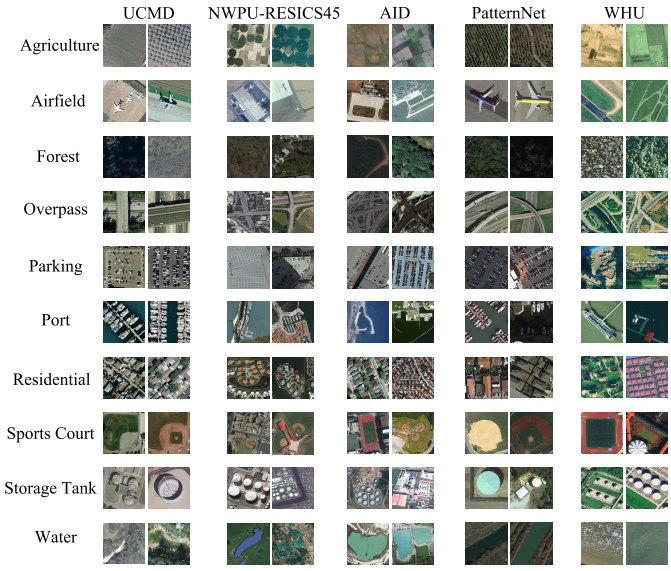


Fig. 6. Example images of ten classes in our dataset for the MTDA task.

where α be defined as $\alpha = (1 - (T/T_{\max})^2) \cdot \tau$, τ represents a hyperparameter. W_F^T denotes the network parameters, $T_{\max}$ represents the total number of training epochs, T represents the current epoch, and α gradually decreases as the training epochs increase.

Then the final loss function can be written as

$$L = L_{\text{cls}} + \lambda_1 L_c + \lambda_2 L_{\text{HFE}} + \gamma L_{\text{PA}} \tag{10}$$

where L_{cls} is the cross entropy loss function [2]; λ_1, λ_2, and γ are the hyperparameters.

In MTDA, the HFPAN leverages the strengths of CNNs for extracting fine-grained features. These features, characterized by their high resolution and specificity, are instrumental for identifying subtle intra-domain variances and fostering accurate domain-specific adjustments. In addition, the MT component of HFPAN focuses on contextual feature extraction. It interprets the broader scene context, ensuring the network grasps overarching patterns essential for cross-domain coherence and abstract feature alignment. The hierarchical feature space created by this dual approach forms the backbone of our adaptation strategy. It enables the network to function across different levels of data representation, methodically aligning features from global to local scales. The outcome is an enhanced capability of the network to transfer knowledge effectively across varied target domains.

IV. Experiments

A. Datasets

Our proposed MTDA method was evaluated on five RS datasets: UC Merced (U), PatternNet (P), AID (A), NWPU-RESISC45 (N), and RSD46-WHU (R). Examples in each dataset are shown in Fig. 6.

The UC Merced (U) dataset includes 2100 RGB images obtained from the United States Geological Survey (USGS) National Map, capturing various regions of the United States. It is classified into 21 classes, with each class containing 100 images.

The PatternNet (P) dataset comprises 800 RGB images covering 38 scene classes. These images are collected from high-resolution Google Earth imagery.

The AID (A) dataset contains 10 000 RGB images collected from Google Earth images taken by aircraft and satellites (e.g., WorldView-2, GeoEye-1, and Landsat 8). It consists of 30 aerial scene classes.

The NWPU-RESISC45 (N) dataset comprises 31 500 images extracted from Google Earth imagery, categorized into 45 scene classes.

The RSD46-WHU (R) dataset contains 117 000 RGB images manually collected from Tianditu and Google Earth. It is categorized into 46 classes, with each class containing 500–3000 images.

We followed the evaluation protocol outlined in [19] and conducted tests on these datasets. Among the datasets, ten classes were shared: agriculture, forest, airfield, overpass, port, storage tank, residential, parking, sports court, and water. Additional details about these datasets can be found in Table I.

B. Parameter Setup

In the experiment, we set the hyperparameters ε and ϕ to 0.4, the learning rate to 0.005, and λ_1, λ_2, and γ to 0.4. The model is trained for 220 epochs, and the batch size is set to 32. All experiments were conducted on a server with an Intel Xeon Silver 4210R processor, NVIDIA Geforce RTX3090 GPU, and 192 GB of DDR4 memory. The Adam optimizer is used to train the proposed decoupling model, and ResNet34 serves as the CNN backbone.

To assess the effectiveness of our proposed HFPAN, we conducted a comparative analysis with state-of-the-art (SOTA) methods in two groups: single-source DA (S3TDA) and multiple-target DA (MTDA). Our objectives were twofold: first, to demonstrate the challenges of extending S3TDA methods to the MTDA setting, and second, to show that our proposed method overcomes the limitations of previous MTDA methods by achieving superior classification performance.

The S3TDA methods we compared include deep adaptation network (DAN) [53], joint adaptation networks (JANs) [54], domain-adversarial neural network (DANN) [36], conditional domain adversarial networks (CDANs) [55], and AMRAN [8]. The MTDA methods we compared include TSAN [19], D-CGCT [56], and AMRAN.

C. Comparison With the SOTA Methods

Table II presents a comparison of the classification results obtained from various benchmark methods and our proposed approach across five DA tasks, considering the single-target (S-T) and target-combined (T-C) settings [33]. The S-T framework is applied to a DA scenario in which the model undergoes fine-tuning and assessment with data from a source and target domain, offering insights into the model's adaptability to a specific domain shift between two discrete domains. In contrast, the T-C framework encompasses a broader scope of DA. Within this framework, the model is trained using labeled data from a single source domain and is

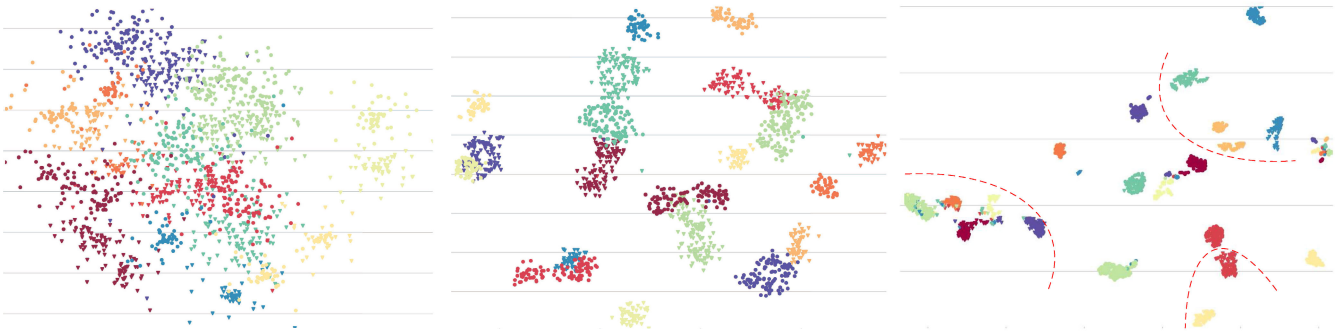
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TABLE I
DETAILED INFORMATION OF FIVE RS SCENE DATASETS

Dataset	AID (A)	RSD46-WHU (R)	UC Merced (U)	NWPU-RESISC45 (N)	PatternNet (P)
Total images	10,000	117,000	2,100	31,500	30,400
Number of classes	31	46	21	45	38
Images per classes	220-420	500-3,000	100	700	800
Resolution(m)	0.5-8	0.5-2	0.3	0.2-30	0.062-4.693
Size	600X600	256X256	256X256	256X256	256X256
Source	Google Earth	Google Earth, Tianditu	USGS	Google Earth	Google Earth, Google Map

TABLE II
ACCURACY OF THE PROPOSED AND BENCHMARK METHODS OVER FIVE DA TASKS

Setting	Single Target	U				A				P				N				R			
		A	P	N	R	U	P	N	R	A	U	N	R	U	A	P	R	U	A	P	N
S-T	DAN	68.3	94.0	62.5	53.4	77.6	82.6	84.2	68.5	66.5	83.4	69.3	52.9	82.9	89.0	92.2	65.9	63.8	69.2	67.7	64.9
	DANN	81.0	93.9	73.7	53.9	92.1	90.1	89.4	77.9	77.9	94.5	75.6	55.7	88.4	92.7	91.0	69.2	72.9	80.9	80.5	78.9
	JAN	75.6	92.5	78.5	62.3	77.8	91.0	88.1	72.2	77.0	90.9	**79.8**	69.3	86.0	89.4	92.0	66.8	70.3	73.3	75.9	76.0
	CDAN	85.5	93.1	75.1	57.0	92.3	97.4	95.0	81.2	77.9	**98.1**	78.5	58.3	96.2	97.1	93.3	84.1	73.3	83.4	81.2	79.7
	AMRAN	82.8	93.5	78.5	60.8	92.8	95.1	94.9	83.5	72.6	93.5	70.5	60.5	94.1	94.8	93.7	83.1	73.8	84.9	80.4	79.8
	HFPAN	**87.6**	**94.5**	**80.2**	**63.5**	**94.6**	**97.8**	**95.8**	**86.0**	**78.3**	94.0	77.3	**69.5**	**96.3**	**97.5**	**94.1**	**85.5**	**74.5**	**85.2**	**82.0**	**80.4**
T-C	DAN	60.6				76.3				55.1				74.0				68.0			
	DANN	63.3				79.7				61.7				76.5				74.8			
	JAN	71.6				78.5				**75.3**				73.0				69.4			
	CDAN	**71.7**				84.1				67.9				79.1				76.6			
	AMRAN	65.7				83.2				69.5				84.2				87.9			
	TSAN	68.1				84.3				60.5				81.9				73.4			
	HFPAN	**71.7**				**85.6**				72.4				**84.3**				**89.3**			



(a) (b) (c)

Fig. 7. Visualization of the HFE Loss for the MTDA task $N \rightarrow \{A, P, R, U\}$. Different shapes represent distinct levels of features, with asterisks indicating fine-grained features and circles representing contextual features. (a) Epoch = 5, (b) epoch = 20 and (c) epoch = 40.

subjected to unlabeled data from various target domains. The evaluation proceeds using a synthesized dataset composed of instances from each target domain, thus challenging the model to demonstrate its generalization capacity to handle complex domain shifts. These configurations play a pivotal role in our experimental setting, allowing for an in-depth evaluation of the HFPAN's adaptability to the diverse degrees of domain complexity encountered in RS image scene classification (RSICS) task.

The results in Table II demonstrate that methods specifically designed for the S-T and T-C settings achieved outstanding performance regarding average classification accuracy. The classification results in Table II reveal two important aspects of MTDA. First, the existing S3TDA approaches encountered difficulties in extending to the MTDA setting. This highlights the challenges in adapting methods designed for single-target scenarios to handle multiple targets simultaneously. Second, as mentioned in Section I, merging different target domains into a single-target domain can compromise the semantic information of each target domain, negatively impacting the discriminative ability of the trained network. Additionally, mixing between multiple domains can lead to feature confusion. On the other hand, the average classification results for the T-C setting were lower than those for the S-T setting.

Our proposed method, HFPAN, achieved the highest average classification accuracy across all settings. Specifically, it achieved an accuracy of 80.6% in the T-C setting and 85.7% in the source-target setting. In terms of S-T methods, HFPAN consistently outperformed other approaches, achieving the highest average accuracy across all five transfer tasks. With

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the exception of the $P \rightarrow U$ and $P \rightarrow N$ transfer tasks, our HFPAN surpassed other DA methods in terms of classification accuracy. As for T-C methods, HFPAN consistently achieved the highest average accuracy of 80.6% across all five transfer tasks. With the exception of the $P \rightarrow A, U, N, R$ transfer task, our HFPAN outperformed other DA methods regarding classification accuracy. Notably, HFPAN exhibited superior performance compared to the second-best approach, AMRAN, with an improvement of 2.5% and 2.6% in the T-C and S-T settings, respectively. Similarly, HFPAN outperformed TSAN by 7.0% in the T-C settings. The success of our method in improving cross-domain classification can be attributed to its consideration of the contextual relationships present in RSI and its hierarchical alignment of fine-grained features. By aligning features from coarse-grained to fine-grained, our proposed model avoids the overfitting of specific feature classes during the alignment process. The hierarchical feature space introduced in our approach plays a crucial role in achieving this improvement. Nevertheless, HFPAN presents certain challenges. The inclusion of multiple alignment stages adds to the model's intricacy, potentially increasing the computational overhead during both training and inference phases. Such complexity necessitates meticulous parameter tuning to ensure a delicate balance between effective alignment and the risk of overfitting, posing a constraint for computationally constrained applications. Moreover, while HFPAN's progressive alignment strategy is tailored for complex, multi-DA scenarios, its efficiency may not be optimal for simpler domain shifts where more straightforward models may be adequate.

In summary, the results unequivocally establish the superior classification accuracy of HFPAN over other DA methods, affirming its effectiveness in tackling the complexities of MTDA. However, the aspects of computational intensity and sensitivity to parameter settings are acknowledged as pivotal points for ongoing enhancement.

D. Ablation Experiment

The ablation experiments were conducted on the MTDA dataset to analyze the proposed DA network. The experimental results are shown in Table III and Figs. 7–11, where ✓means adding the module. Feature visualization effectively assesses the transferability of knowledge from a source domain to multitarget domains, focusing on feature alignment at the domain level and feature discriminability at the class level. Additionally, according to the results presented in Table III, our proposed modules consistently enhance the DA performance of the network.

1) Feature Visualization of the Proposed HFE Loss: Using Uniform Manifold Approximation and Projection (umap), we visualized the feature alignment of these methods on the $N \rightarrow \{U, A, P, R\}$ DA task, as shown in Fig. 7. Our evaluation includes inter and intraclass levels. Fig. 7 presents the visualization of the learned HFE on the $N \rightarrow \{U, A, P, R\}$ DA task. It illustrates the hierarchical feature construction process. With the HFE loss, two HFE clusters are formed in MTDA, but the boundary appears disorganized. In particular, including $L_{\text{HFE-F}}$ and $L_{\text{HFE-G}}$ results in more compact clusters and a clearer boundary between classes. Furthermore, the introduction of the L_{BR} constraint further enhances the effect. Moreover, by incorporating the HFE Loss, the two HFEs for fine-grained and contextual features are constrained by both interclass and intraclass constraints, resulting in even tighter clusters and a more discriminative boundary for MTDA. Additionally, we evaluated the ability and impact of HFE clustering. As illustrated in Fig. 7, through the $L_{\text{HFE-F}}$ and $L_{\text{HFE-G}}$ constraint, the HFE Loss can form fine-grained intraclass clusters for multidomain fine-grained and contextual features. Particularly, the HFE constraint allows each class to cluster more closely, resulting in distinct margins between them. Consequently, HFE significantly aids RSISC in MTDA.

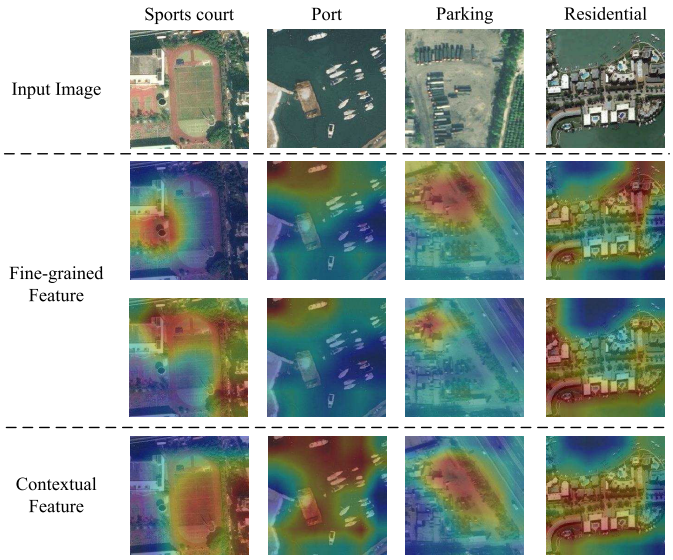


Fig. 8. Attention map visualization results using Grad-Cam on the $N \rightarrow \{U, A, P, R\}$ MTDA task. In the visualizations, regions of interest identified are highlighted in red, while regions considered less important are depicted in blue.

2) Feature Visualization of the Proposed MHPA Loss: Evaluating the transferability of knowledge from the source domain to multitarget domains involves assessing feature alignment (at the domain level). The results of domain feature visualization without using MHPA Loss are presented in Fig. 9, along with a comparison to the AMRAN method. As shown in Fig. 9, neither the AMRAN method nor the method without MHPA Loss performs as well as our proposed method regarding domain feature alignment. The results demonstrate that AMRAN achieved satisfactory domain alignment but struggled to discriminate target representations, as shown in Fig. 9 for the specific target domain. HFPAN (without MHPA Loss) primarily focuses on reducing domain discrepancy, overlooking class-level discriminability. By employing MMD to minimize domain discrepancy in each source and target pair, HFPAN (without MHPA Loss) neglects to explore relationships between different target domains. Conversely, the proposed HFPAN (with MHPA Loss) method encompasses feature alignment within each source and target pair while considering different target domains. It avoids overfitting the model to certain types of features during the alignment process. As a result, the trained model exhibits satisfactory performance in all target domains. The proposed

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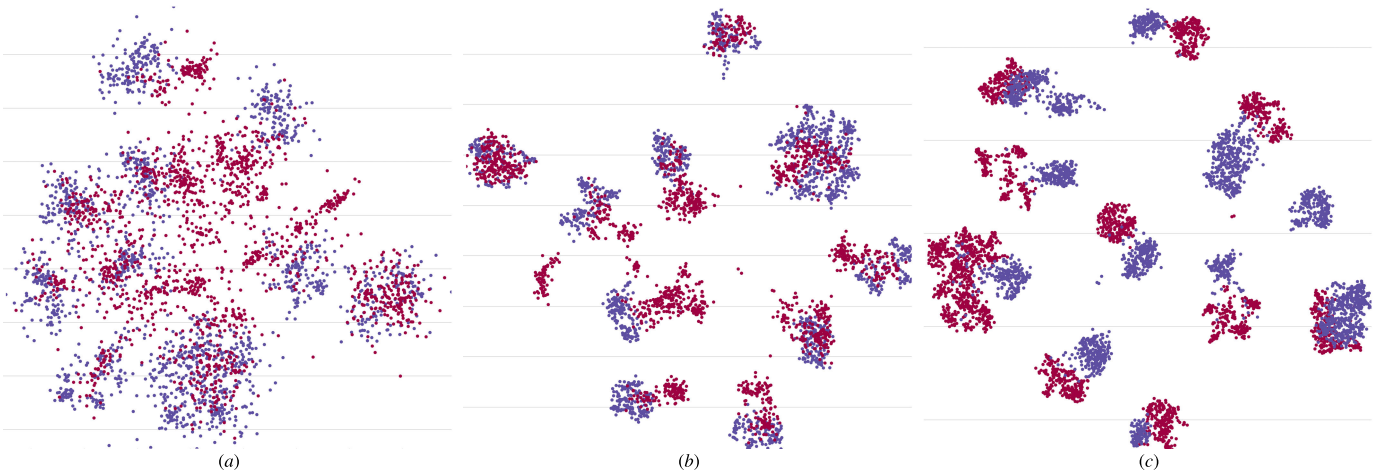


Fig. 9. Feature alignment visualizations on the $N \rightarrow \{U, A, P, R\}$ MTDA task are depicted at the domain level. Blue circles denote samples from the source domain, while red circles denote samples from the combined target domain. (a) Representations of HFPAN without MHPA Loss. (b) Representations of HFPAN with MHPA Loss. (c) Representations of AMRAN.

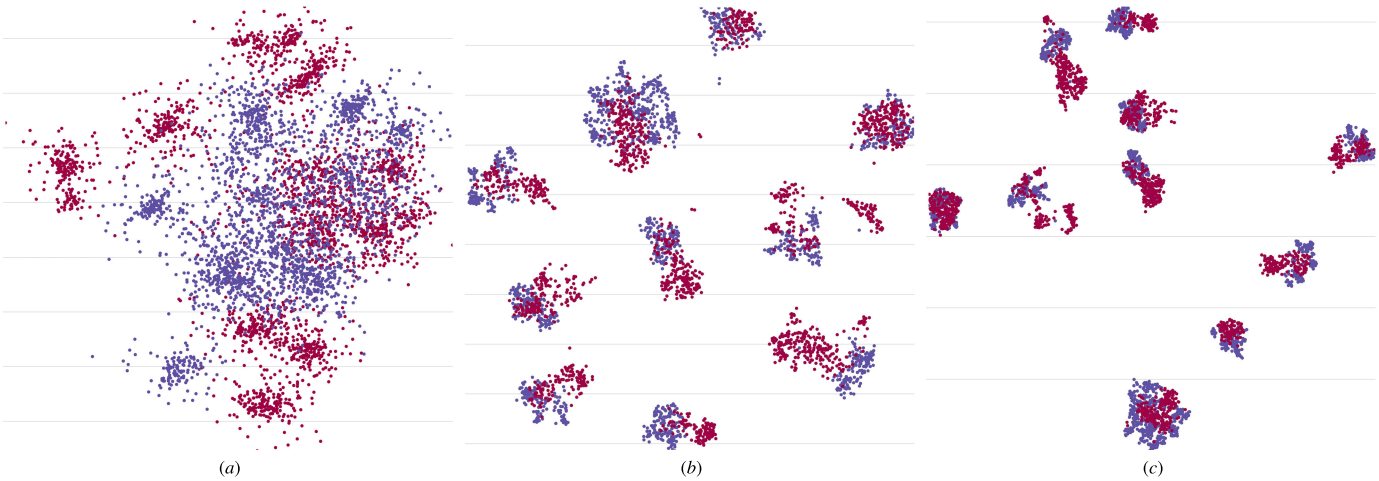


Fig. 10. Visualizations of fine-grained feature alignment on the $N \rightarrow \{U, A, P, R\}$ MTDA task are presented at the domain level. Blue circles denote samples from the source domain, while red circles denote samples from the combined target domain. (a) Representations of HFPAN (epoch =10). (b) Representations of HFPAN (epoch =80). (c) Representations of HFPAN (epoch =150).

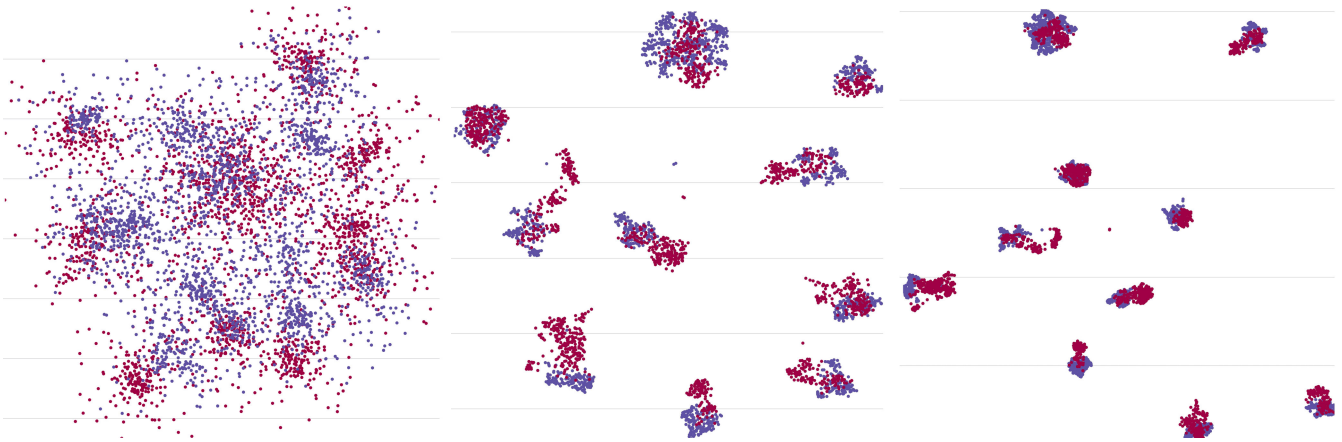
HFPAN (with MHPA Loss) method demonstrates remarkable discriminability in extracting representations for the target domain.

3) Feature Visualization of Different Hierarchical Features: We visualize the impact of the MHPA loss function during the training process through Figs. 10 and 11, illustrating the hierarchical feature alignment process with the progressive loss function. These figures demonstrate that the progressive loss function initially aligns the contextual features of RSI, followed by the fine-grained features, enabling the network to hierarchically align features from easy to difficult. Specifically, in Figs. 10(b) and 11(b), it can be observed that the model exhibits superior alignment performance for contextual features compared to fine-grained features during the mid-stage of training. Furthermore, in Figs. 10(c) and 11(c), toward the end of the training, the model achieves commendable alignment for both contextual and fine-grained features.

4) Attention Map: Fig. 8 presents the attention maps obtained from the proposed HFPAN using Grad-CAM. These attention maps depict the fine-grained and contextual features of the DA task $N \rightarrow \{U, A, P, R\}$. The maps highlight the regions of interest identified by the trained models, with the selected regions displayed in red. We randomly selected input images from three classes: sports court, port, parking, and residential, from the target domains A, P, R, and U. Specifically, our approach accurately focuses on the relevant regions in the selected classes. These visualization results demonstrate the suitability of our proposed method for the multitarget DA task as it effectively captures essential semantic information and improves the discriminative ability in complex scenarios.

5) Parameter Analysis: Fig. 12 depicts the impact of different parameters in the proposed HFPAN model during the training process. Specifically, parameters lambda1, lambda2, and gamma are varied within the range of (0.1, 0.5). As observed from Fig. 12, the testing accuracy of the model exhibits a decreasing trend as lambda1, lambda2, and gamma increase. This observation suggests that the model is more susceptible to changes in lambda2 while being relatively less influenced by variations in lambda1 and gamma.

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(a) (b) (c)

Fig. 11. Visualizations of contextual feature alignment on the $N \rightarrow \{U, A, P, R\}$ MTDA task are presented at the domain level. Blue circles denote samples from the source domain, while red circles denote samples from the combined target domain. (a) Representations of HFPAN (epoch = 10). (b) Representations of HFPAN (epoch = 80). (c) Representations of HFPAN (epoch = 150).

TABLE III

ABLATION EXPERIMENT RESULTS OF THE PROPOSED HFPAN. ✓MEANS ADDING THE MODULE

FCIE	HFE Loss	MHPA Loss	$U \rightarrow \{A, P, N, R\}$	$A \rightarrow \{U, P, N, R\}$	$P \rightarrow \{A, U, N, R\}$	$N \rightarrow \{U, A, P, R\}$	$R \rightarrow \{U, A, P, N\}$
✓			66.4	82.6	69.1	81.3	85.9
✓	✓		68.2	84.5	70.3	83.1	87.4
✓	✓	✓	**71.7**	**85.6**	**72.4**	**84.3**	**89.3**

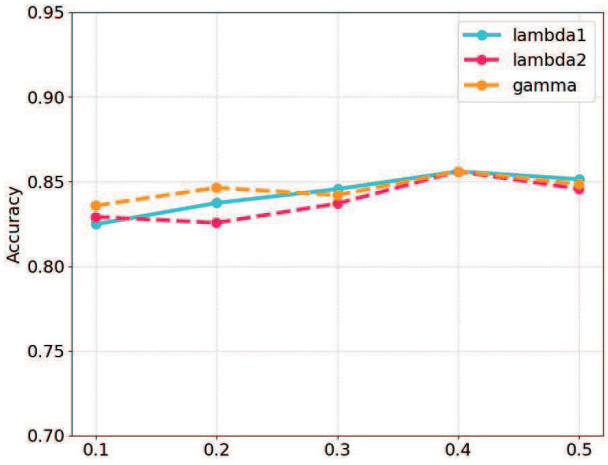


Fig. 12. Accuracy of our method is evaluated on the MTDA task $A \rightarrow \{U, P, N, R\}$, considering different hyperparameters.

V. CONCLUSION

This article proposes the HFPAN for RSISC in MTDA. Our approach has been extensively tested on five RS classification datasets, demonstrating its effectiveness. It is verified that the FCIE can learn fine-grained and contextual features for the MTDA classification task. Moreover, it is proved that the proposed HFE framework can maintain interclass and intraclass feature embedding effectively for multidomain features. Finally, the combination of the HFE framework and the MHPA loss function enhances the ability to minimize differences in MTDA from coarse to fine granularity.

In the future, we intend to extend the application of the HFPAN to a broader array of complex and diverse RS datasets, with a particular focus on partial MTDA scenarios. It will rigorously assess our method's practical utility and real-world efficacy. Additionally, we will explore DA methods without source domain data, aiming to reduce the model's reliance on supervised information in DA tasks.

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